Chapter 5. Association and Independence of Two Variables

Pearson and Spearman correlation, Kendall's tau, permutation tests of independence

Contents

5.1 Pearson correlation **2**

5.2 Spearman's rank correlation **2**

5.3 Kendall's tau **3**

5.4 Selected exercises **4**

5.1 Pearson correlation

$\rho = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y} \in [-1,1]$ measures *linear* association; $\rho = 0$ implies independence under bivariate normality only (e.g. $Y = X^2$, X symmetric, has $\rho = 0$). The sample correlation is $r = S_{XY}/\sqrt{S_{XX}S_{YY}}$ and the LS slope $\hat\beta = S_{XY}/S_{XX} = r\sqrt{S_{YY}/S_{XX}}$; since $\beta = \rho\,\sigma_Y/\sigma_X$, testing $\beta = 0$ and $\rho = 0$ coincide.

Proposition 5.1

If (X_i, Y_i) are i.i.d. bivariate normal with $\rho = 0$, then $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2}$.

Proof. Given the X's, the Y_i follow the regression model with $\beta = 0$, so $\hat\beta/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$. With $SSE = S_{YY}(1-r^2)$ and $\widehat{\mathrm{se}}^2 = SSE/((n-2)S_{XX})$, $\hat\beta/\widehat{\mathrm{se}} = r\sqrt{(n-2)/(1-r^2)}$; the conditional law is free of the X's. ■

Permutation test of independence. Under $H_0 : X \perp Y$, shuffling the Y's among the fixed X's preserves the joint law, giving an exact test with any statistic, e.g. r (sensitive to linear association). H_0 is independence, not $\rho = 0$.

Proposition 5.2

Over all $n!$ permutations of $(Y_1, \ldots, Y_n)$ with the X's fixed, the permutation distribution of r has mean 0 and variance $1/(n-1)$, whatever the data.

Proof. Let $a_i = X_i - \bar X$, $b_i = Y_i - \bar Y$, π uniform: $r = \sum_i a_i b_{\pi(i)}/\sqrt{S_{XX}S_{YY}}$, $Eb_{\pi(i)} = 0$, $\mathrm{Var}\,b_{\pi(i)} = S_{YY}/n$, $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = -S_{YY}/(n(n-1))$ $(i \neq j)$. With $\sum_{i\neq j} a_i a_j = -S_{XX}$,

$$\mathrm{Var}\left(\sum a_i b_{\pi(i)}\right) = \frac{S_{XX}S_{YY}}{n} + \frac{S_{XX}S_{YY}}{n(n-1)} = \frac{S_{XX}S_{YY}}{n-1}.$$

■

With a finite-population CLT, $\sqrt{n-1}\,r \dot\sim N(0,1)$ under permutation: a distribution-free Proposition 5.1.

5.2 Spearman's rank correlation

$r_s = \mathrm{Corr}(R,S)$, the Pearson correlation of the ranks R_i, S_i. It measures *monotone* association, is invariant under increasing transformations and robust to outliers.

Proposition 5.3

Without ties, $r_s = 1 - \dfrac{6\sum_i (R_i - S_i)^2}{n(n^2-1)}$.

Proof. Both rank vectors have mean $\frac{n+1}{2}$ and $\sum(R_i - \bar R)^2 = \frac{n(n^2-1)}{12}$. Expanding $\sum(R_i - S_i)^2$ gives $\sum(R_i - \bar R)(S_i - \bar S) = \frac{n(n^2-1)}{12} - \frac{1}{2}\sum(R_i - S_i)^2$; divide by $\frac{n(n^2-1)}{12}$. ■

The permutation test with r_s tests independence against monotone association; Proposition 5.2 applies verbatim.

5.3 Kendall's tau

Motivation. Kendall's tau counts concordant minus discordant pairs: monotone association from ranks only.

A pair of observations $i < j$ is **concordant** if $(X_i - X_j)(Y_i - Y_j) > 0$ and **discordant** if < 0. Let $U_{ij} = \operatorname{sgn}[(X_i - X_j)(Y_i - Y_j)]$.

Definition (Kendall's tau)

Sample: $\tau_n = \binom{n}{2}^{-1} \sum_{i<j} U_{ij} = (N_c - N_d)/\binom{n}{2}$. Population: for independent copies $(X, Y), (X', Y')$, $\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\}$.

τ_n is unbiased for τ; unlike r_s, it has a clear population counterpart.

Example 5.1

$x = (1, 2, 3, 4, 5)$, $y = (2, 1, 4, 3, 5)$: only pairs $(1, 2)$, $(3, 4)$ are discordant, so $\tau_n = (8-2)/10 = 0.6$. Rank differences $(-1, 1, -1, 1, 0)$ give $r_s = 1 - 6 \cdot 4/120 = 0.8$, equal to r (the data are their own ranks).

Proposition 5.4 (Normal case)

If (X, Y) is bivariate normal with correlation ρ, then $\tau = \frac{2}{\pi} \arcsin \rho$.

Proof [Supplement]. $(U, V) = (X - X', Y - Y')$ is centered normal with correlation ρ: $U = Z_1$, $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$. In polar coordinates $(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$, Θ uniform, and with $\rho = \sin\phi$, $V = R\sin(\Theta + \phi)$. $\cos\Theta$ and $\sin(\Theta + \phi)$ share a sign on total angle $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\phi}{\pi}$ and $\tau = \frac{2\phi}{\pi}$. ■

Hence $\hat{\rho} = \sin(\pi\tau_n/2)$ is a robust estimator of a normal correlation.

Proposition 5.5

If $X \perp Y$ with continuous margins, then $E\tau_n = 0$ and $\operatorname{Var}\tau_n = \dfrac{2(2n+5)}{9n(n-1)}$. Moreover $\tau_n / \sqrt{\operatorname{Var}\tau_n} \xrightarrow{d} N(0, 1)$.

Proof [Supplement]. $U_{ij} = \pm 1$ with probability $\frac{1}{2}$, $\operatorname{Var} U_{ij} = 1$; pairs with no common index are independent. For a shared index, $E(U_{12}U_{13}) = \left(E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\right)^2 = \left(E(2V-1)^2\right)^2 = \frac{1}{9}$, V uniform. With $\binom{n}{2} \cdot 2(n-2)$ ordered pairs of pairs sharing one index,

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2}\Big(1 + \frac{2(n-2)}{9}\Big) = \binom{n}{2}\frac{2n+5}{9},$$

divide by $\binom{n}{2}^2$. Asymptotic normality: Hoeffding's U-statistic theory. ■

Under independence the exact null law of τ_n is its permutation distribution (Kendall's tables without ties). Three ways to test independence:

(i) a (random) permutation test;

(ii) Kendall's exact null distribution;

(iii) the normal approximation of Proposition 5.5, good from $n \approx 10$ except in extreme tails. Ties give $U_{ij} = 0$; software reports $\tau_b = (N_c - N_d)/\sqrt{(N_0 - T_X)(N_0 - T_Y)}$, $N_0 = \binom{n}{2}$, T_X, T_Y the pairs tied in X or Y. With ties only the permutation test stays exact.

Which measure? Bivariate normal: the t-test of r. Heavy-tailed margins with linear dependence: r_s, τ_n are more powerful. Monotone nonlinear: r_s or τ_n. All three can be near zero under strong non-monotone dependence (e.g. $Y = \sin(3X) + \epsilon$).

5.4 Selected exercises

Motivation. Shuffling in regression, a dependent pair with $\rho = \tau = 0$, and Kendall's null law via inversions.

Exercise 5.2 (When shuffling is justified)

Let $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ with (X_i, ϵ_i) i.i.d., X_i independent of ϵ_i, $0 < \operatorname{Var} X_i < \infty$ and $\operatorname{Var} \epsilon_i < \infty$. Show that (X_1, Y_1) and (X_1, Y_2) have the same joint distribution if $\beta_1 = 0$, and different joint distributions if $\beta_1 \neq 0$.

Solution.

If $\beta_1 = 0$: $(X_1, Y_1) = (X_1, \beta_0 + \epsilon_1)$ and $(X_1, Y_2) = (X_1, \beta_0 + \epsilon_2)$, with X_1 independent of ϵ_1 and of ϵ_2, so both have the product law of X and $\beta_0 + \epsilon$.

If $\beta_1 \neq 0$: $\operatorname{Cov}(X_1, Y_1) = \beta_1 \operatorname{Var} X_1 \neq 0$ while $\operatorname{Cov}(X_1, Y_2) = 0$ ($X_1 \perp (X_2, \epsilon_2)$), so the laws differ.

So shuffling Y among fixed X's preserves the joint law exactly when $\beta_1 = 0$, the null of the permutation test. ■

Exercise 5.5 (Dependence with $\rho = \tau = 0$)

Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Show that X and Y are dependent but $\rho = 0$ and $\tau = 0$.

Solution [Supplement].

Dependence. $Y < \frac{1}{4}$ iff $|X| < \frac{1}{2}$, so $P(X > \frac{1}{2}, Y < \frac{1}{4}) = 0$, while $P(X > \frac{1}{2})P(Y < \frac{1}{4}) = \frac{1}{4} \cdot \frac{1}{2} \neq 0$.

Pearson. $EX = EX^3 = 0$ by symmetry, so $\operatorname{Cov}(X, Y) = EX^3 - EX\, EX^2 = 0$.

Kendall. For an independent copy (X', Y'),

$$(X - X')(Y - Y') = (X - X')(X^2 - X'^2) = (X - X')^2(X + X').$$

Since $P(X = X') = 0$, its sign is a.s. that of $X + X'$, continuous and symmetric about 0; so $\tau = 0$.

Hence $E\tau_n = 0$ as under independence: r, r_s, τ_n have little power against U-shaped dependence. ■

Exercise 5.6 (Kendall's null distribution via inversions)

Assume no ties. List the pairs in increasing order of X and let $\pi(k)$ be the rank of the Y-value of the kth pair, so π is a permutation of $\{1, \dots, n\}$. Let N_d be the number of discordant pairs.
(a) Show that $N_d = \#\{j < k : \pi(j) > \pi(k)\}$ and $\tau_n = 1 - 4N_d/(n(n-1))$.
(b) Let $I_k = \#\{j < k : \pi(j) > \pi(k)\}$. Show that if π is uniform on the $n!$ permutations, then $I_1, \dots, I_n$ are independent and I_k is uniform on $\{0, \dots, k-1\}$.
(c) Deduce $E\tau_n = 0$ and $\operatorname{Var} \tau_n = 2(2n+5)/(9n(n-1))$ under independence of X and Y.

Solution [Supplement].

(a) For $j < k$ in X-order, the pair is discordant iff $\pi(j) > \pi(k)$. Without ties $N_c + N_d = \binom{n}{2}$, so $\tau_n = (\binom{n}{2} - 2N_d)/\binom{n}{2} = 1 - 4N_d/(n(n-1))$.

(b)

Step 1. $\pi \mapsto (I_1, \dots, I_n)$ maps the $n!$ permutations into $\prod_{k=1}^{n}\{0, \dots, k-1\}$, also of size $n!$; it suffices to show injectivity.

Step 2. Recover π backwards: if $\pi(k+1), \dots, \pi(n)$ are known, so is $V_k = \{\pi(1), \dots, \pi(k)\}$, and exactly I_k elements of V_k exceed $\pi(k)$, so $\pi(k)$ is the $(k - I_k)$th smallest of V_k. Start at $k = n$, $V_n = \{1, \dots, n\}$.

Step 3. So the map is a bijection, and uniform π gives a uniform point of the product: the I_k are independent, I_k uniform on $\{0, \dots, k-1\}$.

(c)

Step 1. Under independence the Y's of the X-ordered pairs are i.i.d. continuous given the X's, so π is uniform.

Step 2. $N_d = \sum_k I_k$, $EI_k = \frac{k-1}{2}$, $\operatorname{Var} I_k = \frac{k^2-1}{12}$, so $EN_d = \frac{n(n-1)}{4}$, $E\tau_n = 0$, and

$$\operatorname{Var} N_d = \frac{1}{12}\sum_{k=1}^{n}(k^2-1) = \frac{1}{12}\Big(\frac{n(n+1)(2n+1)}{6} - n\Big) = \frac{n(n-1)(2n+5)}{72}.$$

Step 3. Hence $\operatorname{Var}\tau_n = \frac{16}{n^2(n-1)^2}\cdot\frac{n(n-1)(2n+5)}{72} = \frac{2(2n+5)}{9n(n-1)}$, as in Proposition 5.5.

The exact null law follows from

$$Eq^{N_d} = \prod_{k=1}^{n}\frac{1+q+\cdots+q^{k-1}}{k}.$$

and asymptotic normality from Lyapunov: $\sum_k E|I_k - EI_k|^3 \le \sum_k k^3 = O(n^4)$ while $(\operatorname{Var} N_d)^{3/2} \asymp n^{9/2}$. ■